

Mathematics

A Cultural Heritage

Lecture 8 (part 3)

Famous Theorems

Course Topics

*Funny Side of
Mathematics*

Mathematics,
literatures, movies,
and more

**Mathematics
and the Arts**

**Mathematics
and Education**

Two mathematics
cultures and a few
mathematical lives

Numbers

Numbers
at work

Our Obsessions
with Numbers

**Mathematics
and Society**

Mathematics and Truth
(Philosophical Aspects
of Mathematics)

**THEOREMS (PROBLEMS)
AND PROOFS**

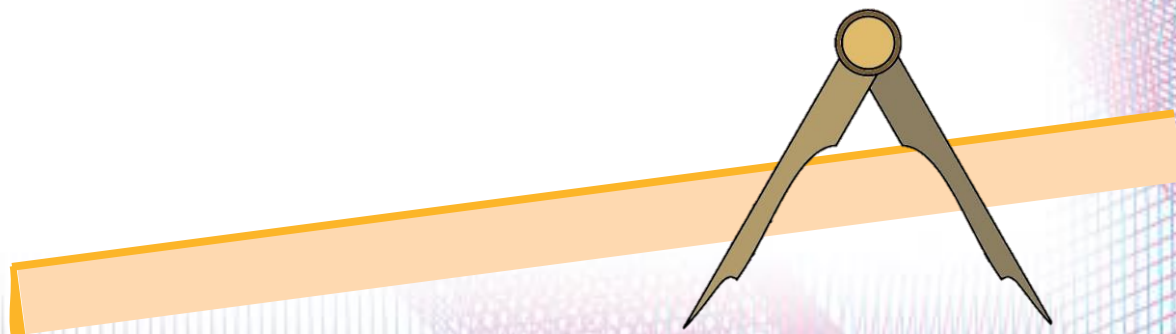
Some Famous Theorems

The Three Greek Problems of Antiquity

The three geometry problems of antiquity:

1. Trisect any given angle
2. Duplicate a cube (double volume)
3. Square a circle (square = area)

by using only a **compass** and a **straight edge/ruler** (that has no markings)



Compass and Straightedge Construction

1. It can be shown (with rigorous mathematics) that, using the compass and a straightedge (ruler), starting with a given length a , the only lengths that can be constructed are formed by finitely many operations using the five basic operations of: $+, -, \times, \div, \sqrt{}$

Compass and Straightedge Construction

1. Without loss of generality we may assume that the given length is 1. Then

- any rational length $\frac{m}{n}$ can be constructed;
- $\sqrt{\frac{m}{n}}$, $\left(\frac{m}{n}\right)^{\frac{1}{4}}$, $\frac{0.2+\sqrt{5}}{\sqrt{1-\sqrt{\frac{3}{7}}}}$, etc., can be constructed.

Doubling the Cube

1. To duplicate a cube, if the length of side of the cube is 1 , then the length x of a side of the bigger cube must satisfy $x^3 = 2 \times 1^3 = 2$, i.e., $x = \sqrt[3]{2}$
2. It can be proved (using **abstract algebra**) that $\sqrt[3]{2}$ cannot be formed by applying finitely many operations of $+, -, \times, \div, \sqrt{}$ on 1 .
3. This means using compass and straight edge **can never duplicate a cube.**

Impossibility of Construction

1. The proof was obtained by **P.L.Wantzel** in 1837.
2. Using similar idea, he actually proved also that **trisecting an angle is impossible.**

Impossibility of Construction

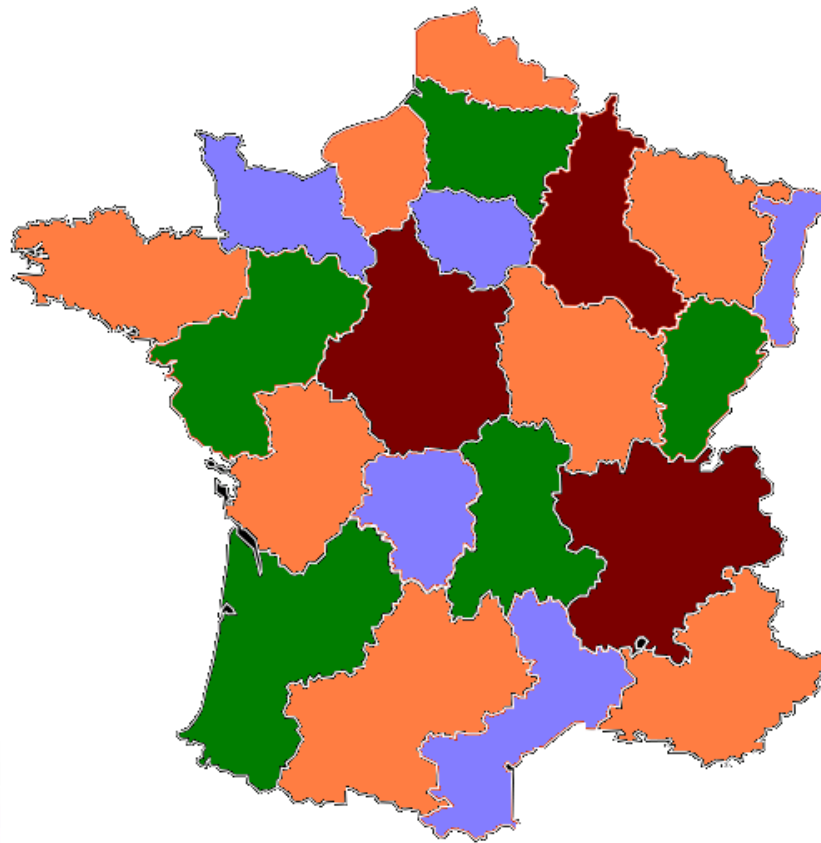
1. To square a circle, if the circle has radius 1 , and the square of the same area has side x , then one must have $x^2 = \pi(1)^2 = \pi$, or $x = \sqrt{\pi}$.
2. In 1882, by showing that both π and $\sqrt{\pi}$ are transcendental numbers, Carl Louis Ferdinand von Lindemann proved that **squaring a circle is impossible.**

Possibility and Impossibility

1. Note that these theorems are about **impossibility**.
2. Utility of impossibility often overlooked.
3. Saves time wasting.
4. Somewhat natural for mathematicians (proof by contradiction)

The Four-Colour Theorem

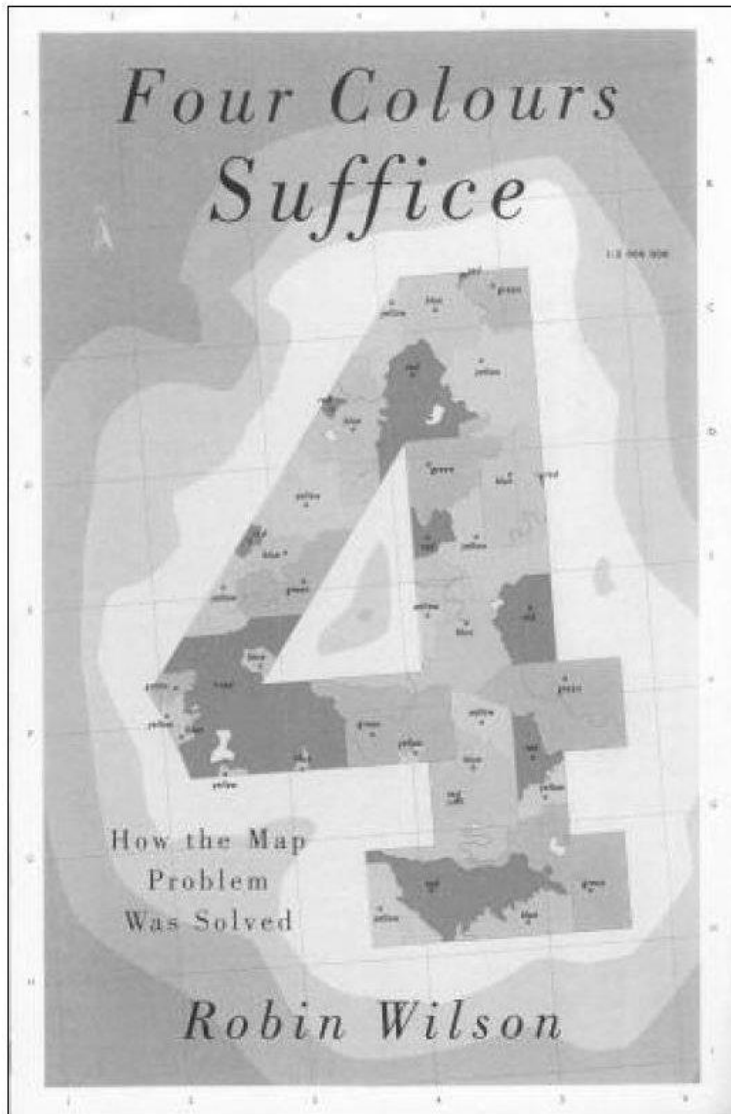
Theorem One needs at most 4 colours to colour any map such that adjacent countries have different colours.



History

1. Conjectured by Englishman **Francis Guthrie** in 1852.
2. Mathematician **Augustus De Morgan** knew the conjecture from Guthrie's brother, and communicated this with other mathematicians.
3. Several failed proofs appeared in late 1800s. Some were considered as correct proofs for some time.
4. It was found out later that these proofs are wrong.
5. **Percy Heawood** proved the 5-colour Theorem in 1890.

1. In 1976, **Kenneth Appel** and **Wolfgang Haken** at the University of Illinois announced that they had proven the theorem.



Kenneth Appel and Wolfgang Haken in the 1970s

The Proof of the 4-Colour Theorem

1. The proof was assisted by using a **computer** to run some algorithms for 1200 hours to check 1476 configurations to exhaust all cases.
2. To this day there is still no human-only proof of the theorem.

A Diophantine Equation

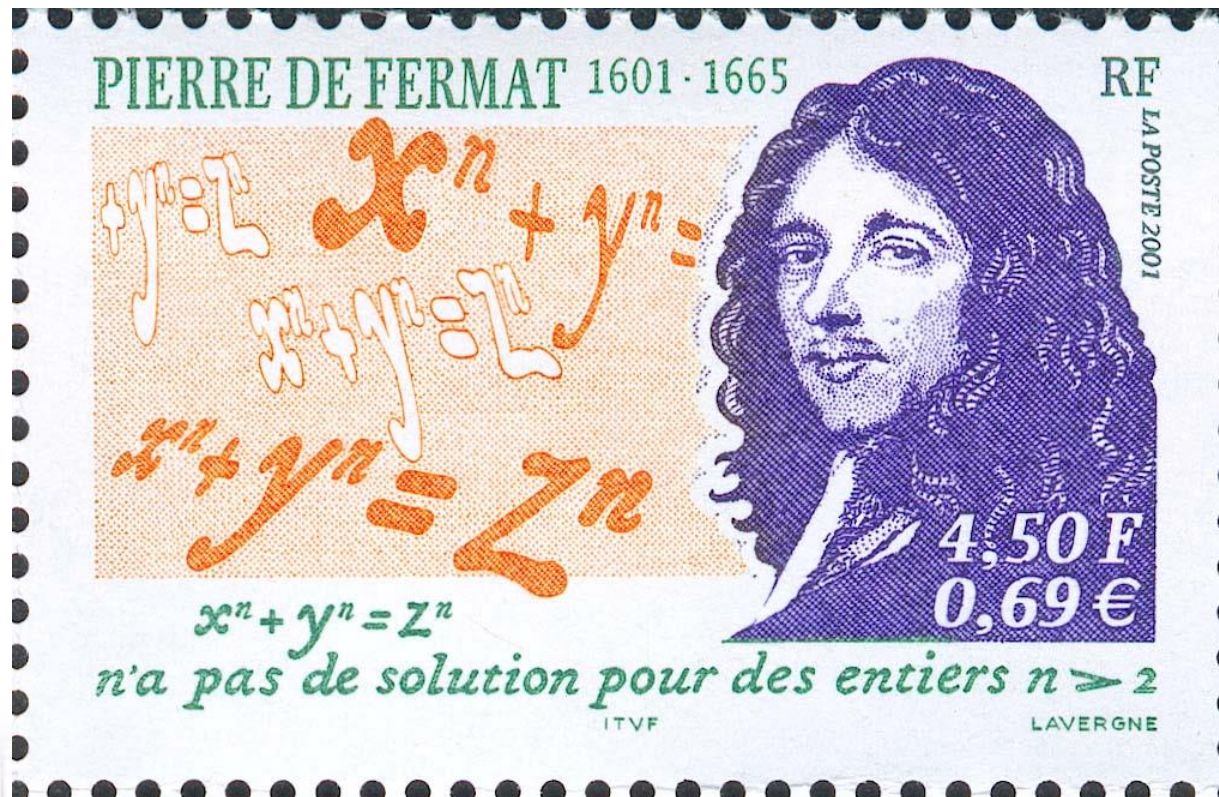
1. Consider the following Diophantine Equation: $x^2 - 1141y^2 = 1$.
2. Obviously, $x = 1, y = 0$ is a pair of integer solution to the equation.
3. **Problem:** Does the equation have any solution in positive integers?
4. **Answer:**
$$1 + 1141x(30693385322765657197397208)^2 = (1036782394157223963237125215)^2,$$

i.e, $x = 1036782394157223963237125215, y = 30693385322765657197397208$ is a solution.

- Does the answer of the previous problem give us any **insight** about the original problem?
- Does the answer help us **understand** the problem?

Fermat's Last Theorem

Theorem The equation $x^n + y^n = z^n$, when n is an integer greater than 2, does not have integer solutions.



Fermat's Last Theorem

1. It was first conjectured in 1637 by **Pierre de Fermat**.

- He wrote in the margin of a copy of *Arithmetica* that he had found a proof, but the margin was too small for him to write down his proof.



Pierre de Fermat (1601-1665)
Source: wikipedia

Fermat's Last Theorem

- Pythagorean triples:

$$x^2 + y^2 = z^2$$

- $x^3 + y^3 = z^3?$ $x^4 + y^4 = z^4?$...?

- Fermat (1637) claimed impossible:

Cubum autem in duos cubos, aut quadratoquadratum in duos quadratoquadratos, et generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est dividere cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non caperet.

It is impossible to separate a cube into two cubes, or a fourth power into two fourth powers, or in general, any power higher than the second, into two like powers. I have discovered a truly marvellous proof of this, which this margin is too narrow to contain.

interuallum numerorum 2. minor autem 1 N. atque ideo maior 1 N. + 2. Oportet itaque 4 N. + 4. triplos esse ad 2. & adhuc superaddere 10. Ter igitur 2. adscitis vnitatibus 10. æquatur 4 N. + 4. & fit 1 N. 3. Erit ergo minor 3. maior 5. & satisfaciunt quæstioni.

εἰ ἐνός, ὁ ἀρα μείζων ἔσται εἰ ἐνός μ² β². δύναιται ἀρα ἀριθμὸς δ' μονάδας δ' τετραπλασίονας εἶναι μ² β². & ἔτι ὑπερέχειν μ² β². τρις ἀρα μονάδας β² μ² β². ἴσαι εἶναι εἰς δ' μονάδας δ' καὶ γίνεται ὁ ἀριθμὸς μ² γ². ἔσται ὁ μὲν ἐλάσσων μ² γ². ὁ δὲ μείζων μ² δ². καὶ ποιῶσι τὸ πρόβλημα.

IN QVÆSTIONEM VII.

CONDITIONIS appositæ eadem ratio est quæ & appositæ præcedenti quæstioni, nil enim aliud requirit quàm vt quadratus interualli numerorum sit minor interuallo quadratorum, & Canones iidem hic etiam locum habebunt, vt manifestum sit.

QVÆSTIO VIII.

PROPOSITVM quadratum diuidere in duos quadratos. Imperatum sit vt 16. diuidatur in duos quadratos. Ponatur primus 1 Q. Oportet igitur 16 - 1 Q. æquales esse quadrato. Fingo quadratum à numeris quotquot libuerit, cum defectu tot vnitatum quod continet latus ipsius 16. esto à 2 N. - 4. ipse igitur quadratus erit 4 Q. + 16. - 16 N. hæc æquabuntur vnitatibus 16 - 1 Q. Communis adiciatur utrimque defectus, & à similibus auferantur similia, fient 5 Q. æquales 16 N. & fit 1 N. $\frac{4}{5}$ Erit igitur alter quadratorum $\frac{16}{5}$. alter verò $\frac{4}{5}$ & vtriusque summa est $\frac{20}{5}$ seu 16. & vterque quadratus est.

ΤΟΝ ὑποταχθέντα τετράγωνον διελὼν εἰς δύο τετραγώνους. ἐπιτεταχθῶ δὴ τὸ 16 διελὼν εἰς δύο τετραγώνους. καὶ τεταχθῶ ὁ πρῶτος δυνάμεις μιας. δύναιται ἀρα μονάδας 16 λείπει δυνάμεις μιας ἴσας 16 τῷ τεταχθῶ. πλάσσω τὸ τετράγωνον ὡς εἰς, ὅσον δὴ ποτε λείπει ποσῶν μ² ὅσον ὅστις ἢ τὸ 16 μ² πλάσσω. ἔστω εἰς β² λείπει μ² δ². αὐτὸς ἀρα ὁ τετράγωνος ἔσται δυνάμεις δ μ² 16 λείπει εἰς 16. ταῦτα ἴσα μονάδας 16 λείπει δυνάμεις μιας. κοινὴ προσκείσθω ἡ λείψας, καὶ ὡς ὁμοίων ὅμοια. δυνάμεις ἀρα εἰ ἴσαι ἀριθμοὶς 16, καὶ γίνεται ὁ ἀριθμὸς 16 πέντε πέντε. ἔσται ὁ μὲν σὺν εἰκοσὶ πέντε. ὁ δὲ μὲν εἰκοσὶ πέντε. & οἱ δύο συντεθέντες ποιῶσι τὸ πρόβλημα.

OBSERVATIO DOMINI PETRI DE FERMAT.

Cubum autem in duos cubes, aut quadratoquadratum in duos quadratoquadratos & generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est diuidere cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non caperet.

QVÆSTIO IX.

VRVSVS oporteat quadratum 16 diuidere in duos quadratos. Ponatur rursus primi latus 1 N. alterius verò quotcunque numerorum cum defectu tot vnitatum, quot constar latus diuidendi. Esto itaque 2 N. - 4. erunt quadrati, hic quidem 1 Q. ille verò 4 Q. + 16. - 16 N. Cæterum volo vtrumque simul æquari vnitatibus 16. Igitur 5 Q. + 16. - 16 N. æquatur vnitatibus 16. & fit 1 N. $\frac{4}{5}$ erit

ΕΣΤΩ δὴ πάλιν τὸν 16 τετράγωνον διελὼν εἰς δύο τετραγώνους. τεταχθῶ πάλιν ἢ τὸ πρῶτον πλάσσω εἰς ἐνός, ἢ τὸν ἑνὸς εἰς ὅσον δὴ ποτε λείπει μ² ὅσον ὅστις ἢ τὸ διαγρημὸς πλάσσω. ἔστω δὴ εἰς β² λείπει μ² δ². ἴσονται οἱ τετράγωνοι ὅς μὲν δυνάμεις μιας, ὅς δὲ δυνάμεις δ μ² 16 λείπει εἰς 16. βλάσσω τὸν δὲ ὡς πρὸ συντεθέντας ἴσους εἶναι μ² 16. δυνάμεις ἀρα εἰ μ² 16 λείπει εἰς 16 ἴσαι μ² 16. καὶ γίνεται ὁ ἀριθμὸς 16 πέντε πέντε.

Attempts of Proving FLT

1. **Euler** proved the general case of the theorem for $n = 3$, **Fermat** $n = 4$, **Dirichlet** and **Lagrange** $n = 5$.
2. The cases of n equal other small primes were proved by others later.
3. Up to 1980's, the computer was used to help prove the cases of all (odd) prime $n < 4,000,000$.
4. There were some false proofs in between.

The Successful Proof

1. In 1993, **Andrew Wiles** gave a presentation of his proof of FLT in a series of talks at a conference in Cambridge.



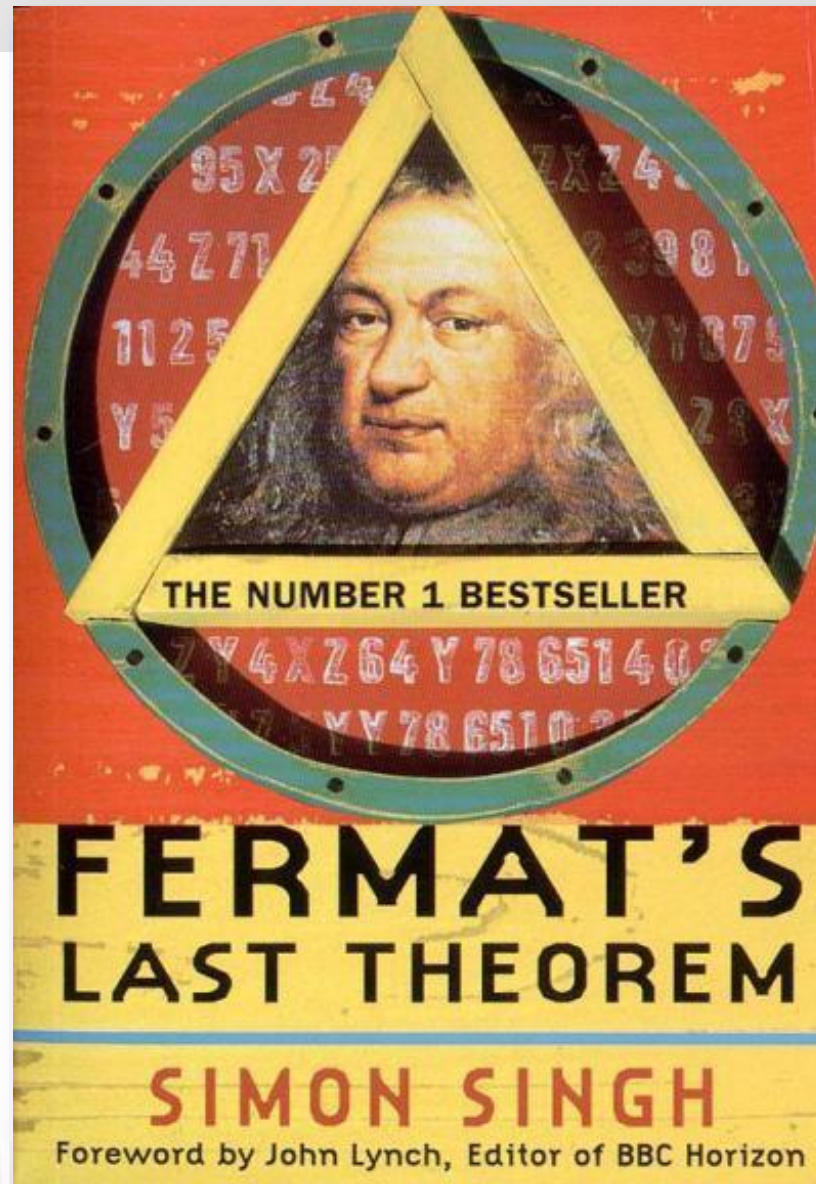
The Successful Proof

1. Wiles's proof was based on the theories of **modular forms** and **elliptic curves**.
2. In August that year, a gap in his proof was discovered.
3. Wiles and his former student **Richard Taylor** worked together to repair his proof. In 1994 they successfully bypassed the gap and completed the proof.

Video: Fermat's Last Theorem



A Good Book on FLT



Basic Idea

- Moving problem is key:

$$x^n + y^n = z^n$$



$$Y^2 = X^3 + AX^2 + BX + C$$



???

Solution

(x, y, z, n)

Elliptic Curve

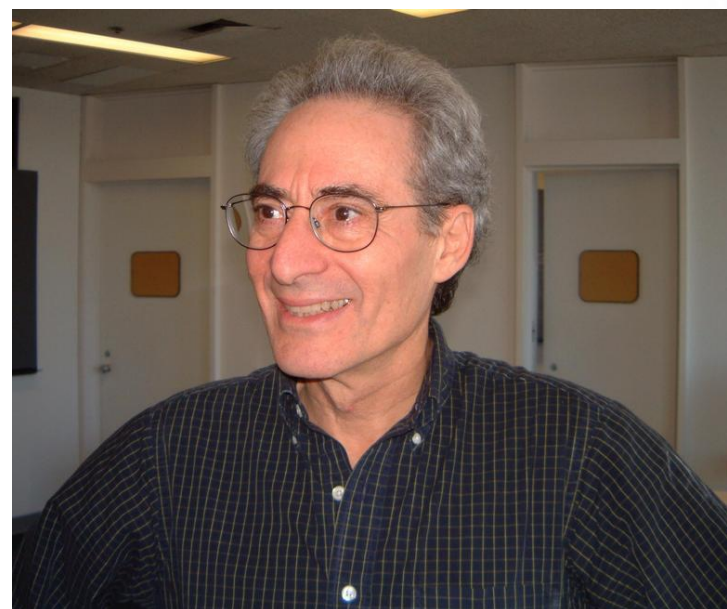
(A, B, C)

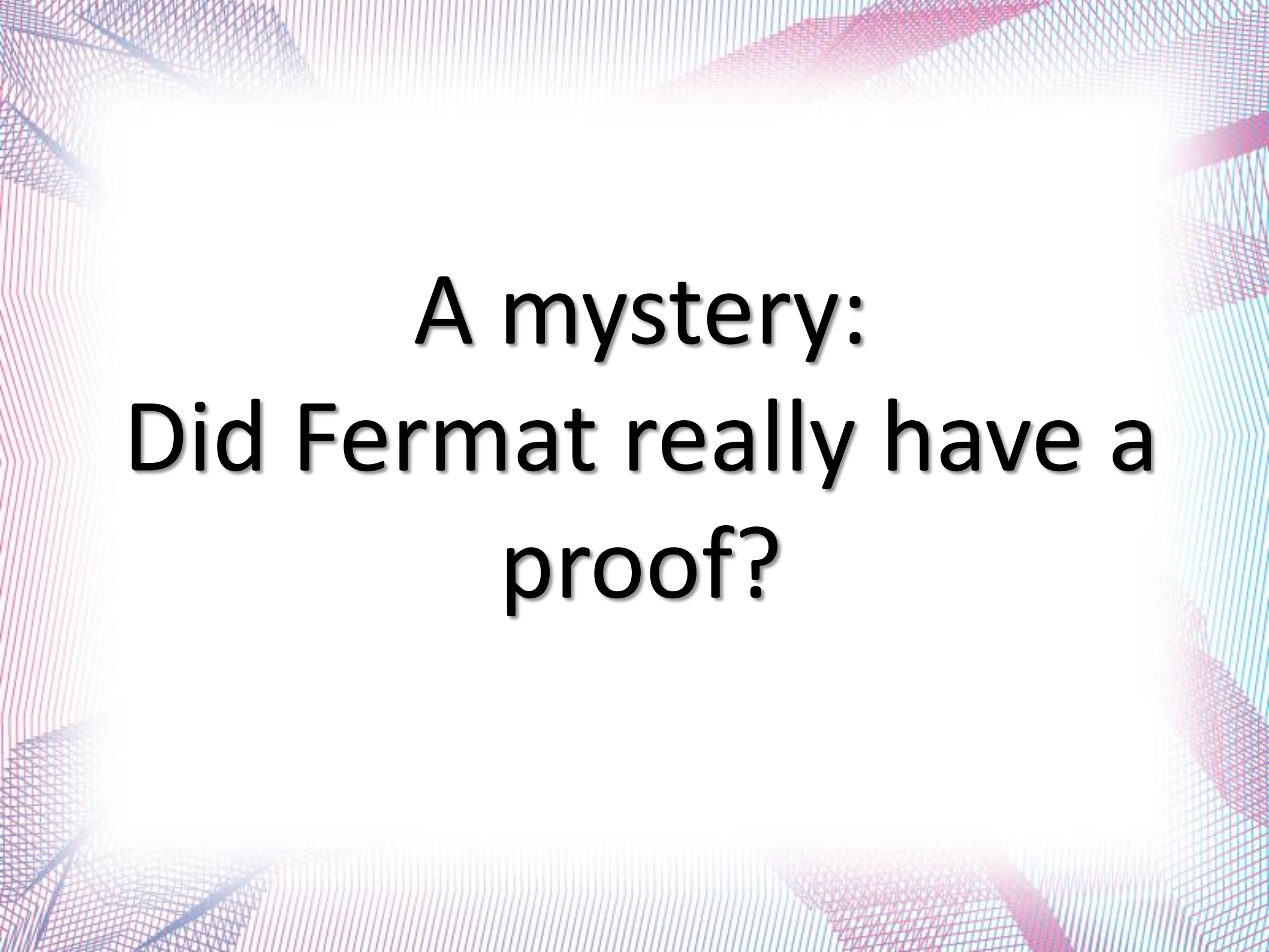
Modular Form

What are Modular Forms?

“Modular forms are functions on the complex plane that are inordinately symmetric. They satisfy so many internal symmetries that their mere existence seem like accidents. But they do exist.”

— Barry Mazur





**A mystery:
Did Fermat really have a
proof?**

Other Significant Problems

Poincaré Conjecture (it is a **theorem** now)

Riemann Hypothesis

P versus NP problem

Goldbach's Conjecture